WEEK 1 · DAY 3

# The Shared Responsibility Model

BaseStack Academy · AWS Cloud Accelerator

Cloud Foundations

# Today's Agenda

The Shared Responsibility Model

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|---|---|---|
| 1 | Overview — Why This Matters | FOUNDATION |
| 2 | Key Concepts — Definitions & Vocabulary | CORE KNOWLEDGE |
| 3 | Deep Dive — Key Concepts Extended | DEEPER LEARNING |
| 4 | How It Works — Step by Step | MECHANICS |
| 5 | Technical Deep Dive | ADVANCED DETAIL |
| 6 | Architecture & Visual Model | VISUAL |
| 7 | Real-World Nigerian Scenario | CONTEXT |
| 8 | Common Mistakes & Exam Traps | EXAM PREP |

# Overview — Why This Matters

The Shared Responsibility Model is heavily tested on the SAA-C03 exam because customer misconfiguration is the #1 cause of cloud breaches. Take the 2019 Capital One breach (100 million records exposed): AWS perfectly secured the physical data centres, but the customer misconfigured their firewalls and access controls. AWS built a secure house, but the customer accidentally left the digital front door wide open.

Many Nigerian organizations fear that moving to the cloud means losing control of their data. As a cloud architect, your job is to explain the reality: while AWS locks the physical data centre doors, the customer holds the only digital keys to their information

## WHY NOW?

Cloud skills are the #1 most in-demand tech capability globally. Understanding this topic is essential for both the SAA-C03 exam and real cloud engineering work.

**Beginner Note**

Do not worry if this feels new — every expert was once a complete beginner. Take notes and ask questions freely.

# Key Concepts — Know These Definitions

Study tip: Say each definition aloud — spoken repetition builds memory 3× faster than reading alone.

### Security OF the Cloud

AWS's responsibility — physical facilities, hardware, virtualisation, global network

### Security IN the Cloud

Customer's responsibility — OS patches, application security, data, identity management

### Managed Services Shift

The more managed the service, the more AWS takes over (Lambda vs EC2 — AWS manages much more for Lambda)

### Shared Controls

Patch management, configuration management, security awareness — both parties share responsibility

### Inherited Controls

Customers inherit AWS's physical and environmental controls — no need to audit the data centre yourself

# Key Concepts (Continued)

## Concepts Summary

All key concepts for this lesson have been covered. Use this slide to review and quiz yourself.

- ✓ Security OF the Cloud
- ✓ Security IN the Cloud
- ✓ Managed Services Shift
- ✓ Shared Controls
- ✓ Inherited Controls

# ⚙️ How It Works — Step by Step (Legal Contract)

1. Think of the Shared Responsibility Model as a contract that clearly delineates ownership. AWS guarantees that the physical data centres are secure — they handle perimeter security at the facility, server hardware, network equipment, and the virtualisation hypervisor layer. If a disk fails in an AWS data centre, AWS immediately replaces it; you never have to touch or worry about the physical hardware.

2. However, everything you deploy ON that infrastructure is your responsibility. If you launch an EC2 instance running an outdated operating system with known vulnerabilities, that is your risk to manage. If you misconfigure an S3 bucket to be publicly accessible when it should be private, AWS will not automatically fix the permissions for you or prevent the resulting data breach.

3. The responsibility line shifts depending on the service type. With EC2 (IaaS), you manage the OS, patching, and applications. With RDS (managed database), AWS manages the underlying OS and database engine patching for you — but you are still responsible for what data is stored, who can access it, and your database firewall rules, network encryption, and password policies.

# ⚙️ Technical Deep Dive

## Understanding the Fundamentals

- The Dividing Line: The Shared Responsibility Model clearly defines who owns what in a cloud environment.

- What AWS Secures (OF the Cloud): The physical facilities, server hardware, network equipment, and the virtualisation layer.

- What You Secure (IN the Cloud): Your data, application code, operating system patches, and firewall configurations.

- The Harsh Reality: If you launch an EC2 instance with vulnerabilities or leave an S3 bucket open to the public, AWS will not fix it for you. You must manage your own risk.

# Architecture & Visual Model

| Core Components | How They Connect |
| --- | --- |

### SECURITY OF THE CLOUD (AWS)

AWS's responsibility — physical facilities, hardware, virtualisation, global network.

### SECURITY IN THE CLOUD (CUSTOMER)

Customer's responsibility — OS patches, application security, data, identity management.

### MANAGED SERVICES SHIFT

Infrastructure (EC2) → More Managed (Lambda)

The more managed the service, the more AWS takes over (Lambda vs EC2...).

SHARED RESPONSIBILITY ZONE

INHERITANCE

### SHARED CONTROLS

Patch management, configuration management, security awareness — both parties share responsibility.

### INHERITED CONTROLS

Customers inherit AWS's physical and environmental controls — no need to audit data centre...

# Real-World Nigerian Scenario

A healthcare startup in Lagos stores patient medical records on Amazon S3. AWS ensures that the physical hard drives storing this data are kept in a locked, access-controlled, and CCTV-monitored facility—that is strictly AWS's responsibility. However, the startup must manage the digital security. They are required to encrypt the data, restrict access using IAM bucket policies so only authorized doctors can read the records, enable S3 access logging for audits, and ensure they do not accidentally make the S3 bucket public to the internet.

### Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?

# ⚠️ Common Mistakes & Exam Traps

● These are the most common exam errors — every one costs marks. Memorise them.

✗ AWS is ALWAYS responsible for the physical infrastructure layer — hardware, facilities, global networking. Customers are ALWAYS responsible for their data classification and IAM access controls.

✗ For RDS: AWS manages the database engine and OS patches. But the customer manages database-level user permissions, the data stored, network access rules, and encryption configuration.

✗ The 'grey area' in exam questions is often about Shared Controls — patch management, configuration management. When in doubt, these are shared between AWS and customer.

# SAA-C03 Practice Question



**Question**

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to The Shared Responsibility Model. Which of the following statements is CORRECT?

**A** AWS's responsibility — physical facilities, hardware, virtualisation, global network

**B** AWS is ALWAYS responsible for the physical infrastructure layer — hardware, facilities, global...

**C** For RDS: AWS manages the database engine and OS patches. But the customer manages database-level...

**D** The 'grey area' in exam questions is often about Shared Controls — patch management,...

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**B** AWS is ALWAYS responsible for the physical infrastructure layer — hardware, facilities, global...

**C** For RDS: AWS manages the database engine and OS patches. But the customer manages database-level...

**D** The 'grey area' in exam questions is often about Shared Controls — patch management,...

# Key Takeaways — What You Learned Today

The Shared Responsibility Model

## Concepts Covered

- ✓ Security OF the Cloud
- ✓ Security IN the Cloud
- ✓ Managed Services Shift
- ✓ Shared Controls
- ✓ Inherited Controls

## Before Next Class

1. Review the concepts above without notes
2. Complete the hands-on console lab
3. Post your progress on LinkedIn / X
4. Review the exam traps from the previous slide
5. Check the Further Resources list

# Further Resources & Reading

Go deeper with these official AWS resources:

https://www.capitalone.com/digital/facts2019/

https://aws.amazon.com/blogs/security/aws-hitrust-csf-certification-is-available-for-customer-inheritance/

Homework: Read at least ONE link above before the next class. Post what you learned on the WhatsApp group.

# BASE STACK

A C A D E M Y

**Week 1 · Day 3 Complete**

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**Next: Hands-on Console Lab →**